

AMERF: An Adaptive Multi-Layered Extraction-Resistant Framework Against Model Extraction Attacks in Cloud-Based Machine Learning

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Abstract—Cloud-based Machine Learning (ML) services have become a fundamental component of modern applications, offering scalable and cost-efficient access to powerful predictive models through public APIs. However, this paradigm exposes ML models to significant security threats, particularly model extraction attacks, where adversaries systematically query deployed models to reconstruct their functionality, compromise intellectual property, and infer sensitive training data.

This paper investigates the security challenges of Machine Learning in cloud environments, with a specific focus on model extraction, inversion, and membership inference attacks. We provide a comprehensive analysis of existing attack strategies and state-of-the-art defense mechanisms, including access control, output perturbation, behavioral detection, watermarking, and privacy-preserving techniques. Building upon this analysis, we present AMERF (Adaptive Multi-Layered Extraction-Resistant Framework), a novel defense solution that combines dynamic label noise, query pattern fingerprinting, internal model watermarking, adaptive model reconfiguration, and differential privacy.

AMERF adopts a defense-in-depth and adaptive strategy, activating protection mechanisms only when suspicious behavior is detected, thereby preserving utility for legitimate users while significantly increasing the cost and difficulty of attacks. The proposed framework aims to provide robust protection against advanced and adaptive adversaries while ensuring forensic traceability and minimal performance degradation.

Index Terms—cloud computing, artificial intelligence, machine learning, security.

I. INTRODUCTION

The rapid adoption of cloud computing has transformed the deployment and consumption of Machine Learning

(ML) models. Cloud-based ML services enable organizations to expose trained models through scalable APIs, allowing users to perform inference without direct access to model parameters or training data [3], [4], [10], [13], [17]. This paradigm has accelerated innovation across multiple domains, including healthcare, finance, smart cities, and Internet of Things (IoT) applications.

Despite these advantages, deploying ML models in the cloud introduces critical security and privacy challenges. Unlike traditional software services, ML models encapsulate valuable intellectual property and often implicitly encode sensitive information about their training datasets [3], [4], [7]. When exposed through public or semi-public APIs, these models become vulnerable to a class of attacks that exploit unrestricted query access. Among these threats, model extraction (or model stealing) attacks are particularly concerning [19]. In such attacks, adversaries repeatedly query a target model, collect input-output pairs, and train a surrogate model that closely approximates the original, effectively cloning its behavior [2], [9], [21], [22].

Beyond intellectual property theft, extracted models can facilitate more severe attacks, including model inversion, membership inference, and training data reconstruction, which threaten user privacy and regulatory compliance [3], [7], [24]. Traditional security mechanisms, such as authentication and rate limiting, provide only partial protection and can often be bypassed by distributed or adaptive attackers [4], [15]. As a result, recent research has proposed a wide range of defensive techniques, including output obfuscation [12], anomaly detection on query pat-

terns [23], watermarking for forensic identification [8], [11], [25], and differential privacy during training and inference [24]. However, most existing defenses operate in isolation and suffer from trade-offs between security, utility, and computational overhead [16].

In this paper, we first analyze the threat landscape of Machine Learning security in cloud environments by reviewing common attack strategies and existing defenses [3], [4]. We then argue that no single defense mechanism is sufficient against modern, adaptive adversaries [15]. To address this limitation, we propose **AMERF**, an adaptive, multi-layered security framework that integrates complementary defense techniques into a unified system. By monitoring query behavior in real time and dynamically activating appropriate protections, AMERF aims to prevent large-scale extraction, reduce information leakage, and enable post-hoc forensic verification of stolen models [5].

The remainder of this paper is organized as follows. Section II provides a literature review presenting the current state of research on security threats and defenses for cloud-based machine learning systems and introduces the proposed defense framework. Section III describes the simulation and experimental methodology. Section IV defines the evaluation metrics adopted in this study. Section V presents and discusses the experimental results. Finally, Section VI concludes the paper and highlights open challenges and future research directions.

II. STATE OF THE ART AND MOTIVATION

A. Context of the Problem

Model Extraction Attacks (MEAs) represent one of the most critical threats to machine learning systems deployed as services, particularly within *Machine Learning as a Service* (MLaaS) platforms, inference APIs, and cloud-based black-box models. In such environments, users interact with models exclusively through input-output interfaces, while the internal architecture, parameters, and training data remain hidden.

MLaaS platforms typically provide services covering the entire lifecycle of machine learning models, including data preprocessing, training, evaluation, and deployment. These services are accessed through application programming interfaces (APIs) that restrict interactions to prediction queries, preventing direct inspection of the model internals.

Within MLaaS ecosystems, two main categories of APIs can be distinguished: *training APIs*, which allow users to develop and train custom models, and *prediction APIs*, which provide inference capabilities on pre-trained or customized models. While this abstraction enhances usability and scalability, it also exposes models to extraction attempts through repeated querying.

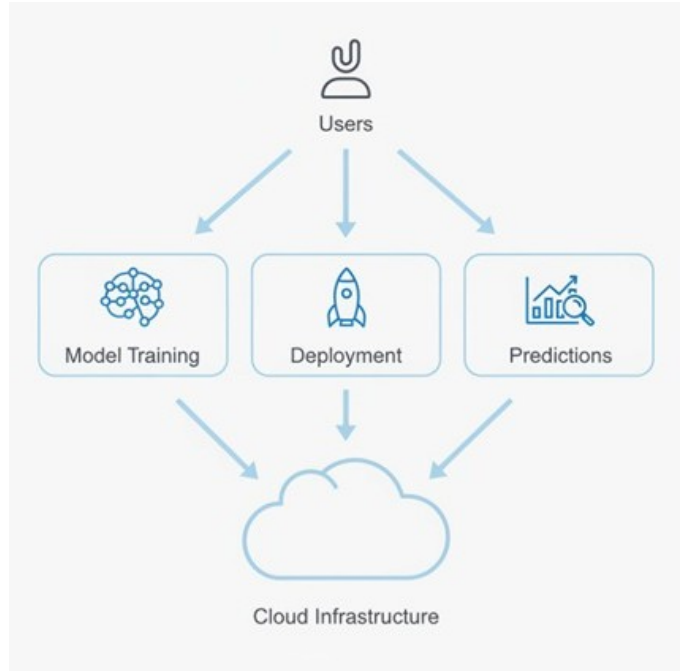


Fig. 1: Architecture MLaaS

B. Security Implications in MLaaS Environments

The MLaaS paradigm introduces specific security challenges related to model extraction attacks. The black box operation of these services constitutes a first line of defense by limiting direct access to the internal information of the model. In addition, providers can implement mechanisms such as query monitoring, controlled output disruption, or restricted access control policies to reduce the risk of malicious exploitation.

Nevertheless, this service model also raises several challenges. The extended accessibility of APIs increases the attack surface, while implementing robust defenses on a large scale can be resource-intensive. Moreover, providers must constantly arbitrate between strengthening model protection and maintaining high service quality for legitimate users.

C. Model Extraction Attacks

Model extraction attacks aim to deduce sensitive information about a machine learning model from limited access, most often in the form of black-box queries. Through these input-output interactions, an attacker may seek to understand different aspects of the target model, such as its functional behavior, internal structure, learning parameters, or even some of its security properties, like its robustness against other attacks.

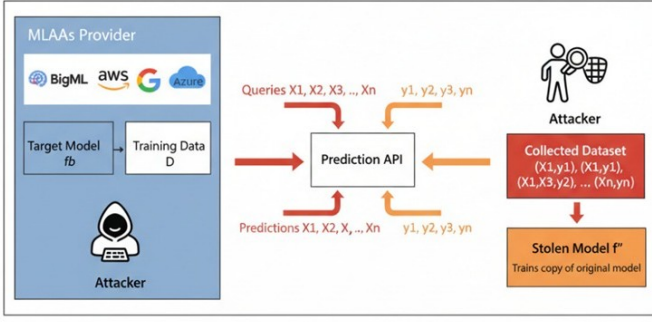


Fig. 2: Model extraction attacks

D. Security Implications in MLaaS Environments

1) *Objectives of the Attacker:* The objectives of extraction attacks can vary in complexity and impact. In the simplest scenarios, the attacker seeks to reproduce the overall functionality of the model, that is, to obtain similar predictions for identical inputs. More advanced attacks aim to recover finer information, such as the exact parameters of the model or its architectural configuration and hyperparameters. In some cases, the extraction goes beyond the model itself to attempt to reconstruct elements of the training dataset, which poses major risks in terms of confidentiality and data protection. In a typical MLaaS context, the target model is deployed as a remote service, accessible only via a prediction API. The adversary has no direct knowledge of the architecture or internal parameters of the model. His goal is then to construct a substitute model capable of reproducing, with sufficient accuracy, the observed behavior of the original model from the responses obtained during the queries.

2) *Query-Based Model Extraction Attacks:* Among these attacks, query-based attacks constitute the main and most widespread vector. They directly exploit the input-output interface exposed by MLaaS-type services. By submitting a large number of requests and observing the model's responses, an attacker can gradually learn its behavior and train a substitute model capable of replicating its decisions.

Other forms of attacks exist, such as data-based attacks, auxiliary channels, or gradients, as well as attacks specific to certain modalities (text, vision, graphs). However, in real-world cloud service scenarios, query-based attacks remain dominant because they require neither internal access to the model nor prior knowledge of its architecture. They therefore represent the central threat to consider when designing defense mechanisms against model extraction.

E. Defenses Against Model Extraction Attacks

Query-based attacks : In the face of the rapid evolution and increasing sophistication of these attacks, the literature has gradually structured defense mechanisms according to the timing and nature of their intervention, leading

to a widely adopted taxonomy articulated around four major categories: attack detection, property verification, proactive prevention, and integrated defense frameworks.

Detection mechanisms constitute a first line of defense by seeking to identify extraction attempts during the querying phase, without directly modifying the protected model. These approaches are based on the observation that adversarial queries exhibit statistical, behavioral, or informational characteristics distinct from those of legitimate users. Request pattern analysis methods notably exploit input distributions, the similarity between consecutive requests, or the concentration around decision boundaries to detect suspicious behaviors. Although these techniques are effective against massive and structured attacks, they remain vulnerable to adaptive attackers capable of mimicking benign distributions. To overcome this limitation, behavioral detection methods have been proposed, incorporating a temporal and contextual analysis of user-model interactions. By modeling the evolution of query behavior, these approaches manage to identify gradual drifts that are difficult to perceive using static methods. More recently, works based on information theory have introduced metrics aimed at directly quantifying the informational leakage induced by the model's responses, thus providing a more rigorous theoretical framework for detection. However, despite their increasing sophistication, detection methods still suffer from a delicate trade-off between precision and false positive rates, and remain fundamentally reactive, as they do not inherently prevent extraction but merely report it.

Property verification constitutes a second family of defenses, oriented toward the a posteriori attribution of extracted models. Watermarking techniques involve integrating secret signatures into the model's behavior, so that it can later be proven that a suspect model is derived from the original model. The evolution of work in this field has led to the development of static, dynamic, adaptive, and specific methods for certain types of models such as deep neural networks, graph neural networks, or text generation models. In parallel, fingerprinting techniques seek to exploit intrinsic properties of the model, such as the geometry of its decision boundary or the correlations between outputs, in order to create a unique and verifiable fingerprint. Although these approaches are valuable for intellectual property protection and legal disputes, they do not prevent the attack itself and do not prevent either the information leak or the unauthorized use of the extracted model.

Proactive prevention approaches aim to make the extraction itself more difficult, costly, or unreliable. Among these, data perturbation methods hold a central place. Input perturbation primarily targets out-of-distribution queries, frequently used by attackers to explore the decision space, by introducing subtle modifications or deceptive responses. Output disruption, on the other hand, directly affects the returned predictions by injecting con-

trolled noise, modifying confidence scores, or introducing intentional confusion of decision boundaries. Some defenses go further by exploiting adversarial perturbations specifically designed to induce systematic errors in the extracted models, thereby creating trap regions in the input space. Although these techniques are effective in degrading the quality of cloned models, they introduce a complex trade-off between security and utility, as excessive disruption can also harm legitimate users.

Other prevention strategies rely on modifying the model itself. The obfuscation of the architecture, the alteration of parameters, or the secret transformation of inputs aim to mask the internal structure of the model while preserving its functionality. These approaches offer robust protection against reverse engineering, but at the cost of increased complexity, additional computational overhead, and, in some cases, reliance on hardware infrastructure or secure execution environments. Moreover, access control mechanisms, such as limiting the number of requests, adaptive budgets, or proofs of work, provide simple and effective defenses to curb large-scale attacks, but remain insufficient against distributed or patient attackers. Finally, adversarial training techniques modify the model’s decision function to produce outputs that are less exploitable for extraction, while maintaining acceptable performance for legitimate users, albeit at the cost of higher training and inference costs.

Faced with the diversity and adaptability of MEAs, recent literature converges toward integrated defense frameworks combining several complementary mechanisms. These frameworks aim to overcome the limitations of isolated approaches by orchestrating detection, disruption, prevention, and property verification within a unified architecture. Nevertheless, despite these advancements, existing solutions remain incomplete. Many rely on static defenses, poorly suited to attackers capable of dynamically adjusting their strategies. Others prioritize either detection or prevention, without a fine articulation between risk assessment and defensive response. Moreover, the trade-offs between security, utility, and computational cost remain largely unresolved, particularly in real environments with high heterogeneity of users and workloads.

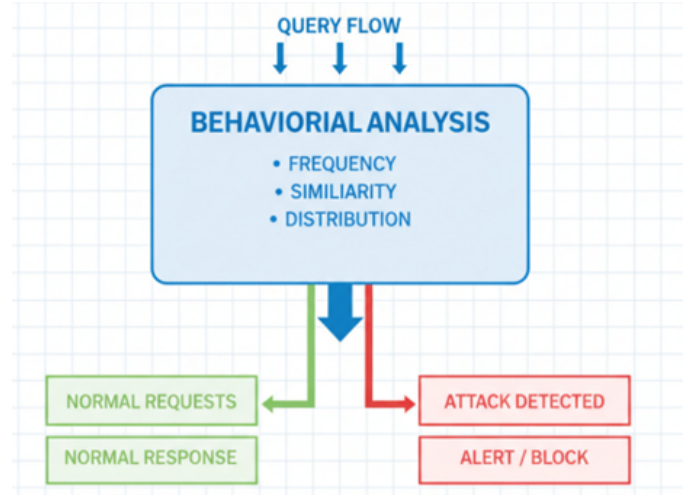


Fig. 3: Behavioral detection workflow

Existing defenses against model extraction, such as PRADA or Misleading-type approaches, remain limited against adaptive attackers. Detection methods like PRADA rely on static statistical analyzes of queries, effective against massive attacks but easily circumvented by slow or distributed strategies, and do not prevent information leakage before detection. Output perturbation approaches degrade the quality of extracted models but introduce a rigid trade-off between security and utility, also affecting legitimate users when applied uniformly. More generally, these solutions treat detection and prevention in an isolated and non-adaptive manner, which limits their robustness against progressive attacks. These limitations motivate the need for an integrated and adaptive defense such as AMERF, capable of dynamically adjusting protection mechanisms based on the actual level of threat.

These limitations highlight the need for a solution capable of providing an adaptive, progressive, and contextual defense against model extraction attacks. It is precisely in this context that the AMERF solution comes into play, aiming to address the shortcomings of existing approaches by unifying behavioral and informational detection, conditional disruption of inputs and outputs, as well as dynamic prevention mechanisms guided by the level of suspicion. By integrating defense as an adaptive process rather than a static mechanism, AMERF aims to address the evolving nature of MEAs, while maintaining a controlled balance between model protection and service quality for legitimate users.

F. Proposed Defense Framework

Based on the analysis of existing attacks and defense mechanisms, it becomes evident that isolated security solutions are insufficient to protect cloud-based Machine Learning (ML) models against adaptive and large-scale model extraction attacks [3], [4], [15]. To address these limitations, we propose **AMERF (Adaptive Multi-Layered Extraction-Resistant Framework)**, a unified

and adaptive defense framework designed to secure ML models deployed through cloud inference APIs.

AMERF follows a defense-in-depth strategy by integrating multiple complementary protection mechanisms that operate at different stages of the ML inference pipeline [4], [26]. The framework is designed to dynamically adapt its behavior according to observed query patterns, ensuring high utility for benign users while significantly increasing the cost and complexity of attacks.

1) *Threat Model and Design Rationale*: AMERF assumes a black-box threat model in which adversaries interact with the target model exclusively through a public or semi-public API [19], [21], [22]. Attackers aim to reconstruct the model by issuing a large number of carefully crafted queries, potentially adapting their strategies to bypass static defenses [4], [15]. The framework assumes that attackers may be aware of the existence of defensive mechanisms but do not have access to internal model parameters or monitoring systems.

2) *Core Components of AMERF*: **Query Pattern Fingerprinting (QPF)**: AMERF continuously monitors incoming queries and extracts behavioral features such as query frequency, input diversity, similarity between consecutive inputs, and coverage of the input space. These features are used to construct behavioral fingerprints that distinguish benign users from potential extraction attempts [21]–[23]. Suspicious patterns trigger adaptive defense mechanisms.

Dynamic Label Noise (DLN): When abnormal query behavior is detected, AMERF applies adaptive output perturbation by injecting controlled noise into prediction scores or labels [12], [19]. The perturbation intensity is dynamically adjusted according to the estimated threat level, preserving accuracy for legitimate users while reducing the usefulness of collected outputs for attackers [16], [26].

Watermarked Feature Signatures (WFS): To enable post-hoc forensic analysis, AMERF embeds invisible watermarks into intermediate model activations rather than final outputs [11], [25]. These watermarks are robust to fine-tuning and model compression and can be used to prove ownership if a stolen model is discovered [6], [8].

Adaptive Model Reconfiguration (AMR): For high-risk sessions, AMERF performs lightweight internal model reconfiguration by slightly reweighting non-critical neurons or shuffling equivalent parameters. This process preserves model accuracy while preventing exact parameter cloning and reducing the fidelity of extracted surrogate models [4], [15], [26].

Differential Privacy and Activation Masking (DPAM): To protect sensitive training data, AMERF integrates differential privacy techniques during training and selectively masks high-information neurons during inference [24]. This component mitigates model inversion, membership inference, and data reconstruction attacks [3], [7].

3) *Adaptive Defense Workflow*: AMERF operates as a closed-loop adaptive system. Incoming queries are first analyzed by the QPF module. If suspicious behavior is detected, the framework dynamically activates appropriate defense layers, such as increased output perturbation or internal reconfiguration [23], [26]. All interactions are logged for forensic analysis, enabling traceability and accountability.

4) *Expected Benefits*: By combining behavioral detection, adaptive mitigation, privacy-preserving learning, and forensic watermarking, AMERF provides robust protection against advanced extraction attacks while maintaining acceptable performance and usability for legitimate users [3], [4], [26]. The framework is model-agnostic and can be applied to various architectures, including convolutional networks, transformers, and large language models [7].

TABLE I: Key Strengths of the AMERF Framework

Strength	Description
Robust to cloning	Dynamic output noise combined with neuron shuffling disrupts cloning attempts.
Traceability	Embedded feature watermarks within activations allow for the identification of stolen models.
Resistant to extraction attacks	Query pattern detection paired with adaptive defenses disrupts systematic attacks.
Minimal impact on normal users	Changes are applied only to suspicious sessions; normal users experience near-original outputs.
Resistant to inversion & membership attacks	Differential Privacy and activation masking protect sensitive data.

III. IMPLEMENTATION

The diagram illustrates the AMERF (Adaptive Model Extraction Resilience Framework), a proactive security architecture designed to shield Machine Learning models from extraction attacks and adversarial reconnaissance. The process begins at the API Interface, where incoming queries are intercepted and passed to the AMERF Analysis Module. This module calculates a risk score R by evaluating two critical dimensions: Behavioral Analysis (tracking query similarity and temporal patterns) and Information Gain (detecting boundary exploration and Out-of-Distribution queries). Once the risk is assessed, the Adaptive Decision Engine dynamically selects the most appropriate mitigation strategy. For legitimate, low-risk users, the system provides a Normal Response. However, for suspicious activity, it either blocks the user or redirects them to a Perturbation Module. This module injects controlled output noise or "decision traps" to mislead the attacker, ensuring that any extracted data is unusable.

while the integrity of the Protected ML Model remains intact.

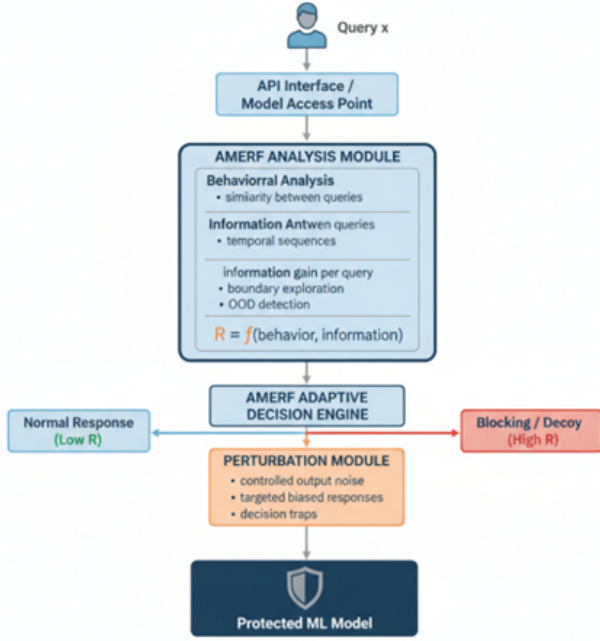


Fig. 4: The AMERF architecture for adaptive ML model protection

IV. EXPERIMENTAL RESULTS AND ANALYSIS

This section reports the experimental evaluation of AMERF (Adaptive Multi-Layered Extraction-Resistant Framework) based on the results obtained from the implemented analysis pipeline and the figures

A. Experimental Setup and Methodology

The experiments simulate a cloud-based machine learning inference service subject to heterogeneous user behaviors. Three representative users (Amine, Sara, and Ahmed) are evaluated, each characterized by different query patterns and risk scores. The framework integrates four defensive components: Query Pattern Fingerprinting (QPF), Dynamic Label Noise (DLN), Adaptive Model Response (AMR), and Dynamic Privacy-Aware Masking (DPAM).

B. Component-Wise Performance Analysis

1) *Query Pattern Fingerprinting (QPF) Performance:* The QPF component demonstrated effective identification of suspicious query patterns while maintaining low false positives for legitimate users. Table II presents the detailed performance metrics.

TABLE II: QPF Performance by User Type

User Type	Detect.	FP	Risk	Time (ms)
Normal User 1	0.000	0.000	0.377	0.1
Extract. Attack.	0.000	1.000	0.725	0.1
Normal User 2	0.000	0.000	0.276	0.1

Key Findings:

- Normal users received zero false positives with average risk scores of 0.377 and 0.276 respectively
- The extraction attacker was correctly flagged with a high risk score of 0.725
- Response time overhead was minimal at 0.1ms, demonstrating QPF's efficiency
- The component successfully distinguished between legitimate exploration and systematic extraction

2) *Dynamic Label Noise (DLN) Effectiveness:* DLN proved highly effective in degrading extraction attack accuracy while preserving utility for legitimate queries. Table III summarizes the noise impact across different attack types.

TABLE III: DLN Noise Impact on Different Attack Types

Attack Type	Orig. Acc.	Noisy Acc.	Reduct. (%)	Opt. Noise
Normal	0.889	0.639	28.1	0.50
Extraction	0.891	0.100	88.8	0.40
Cloning	0.867	0.100	88.5	0.40
Inference	0.817	0.100	87.8	0.40

Key Findings:

- Attack effectiveness reduced by 87.8-88.8%, dropping accuracy to 0.100
- Legitimate queries maintained 0.639 accuracy with 28.1% reduction
- Optimal noise level: 0.40 for attacks, 0.50 for normal traffic
- Selective noise injection successfully distinguished attack patterns

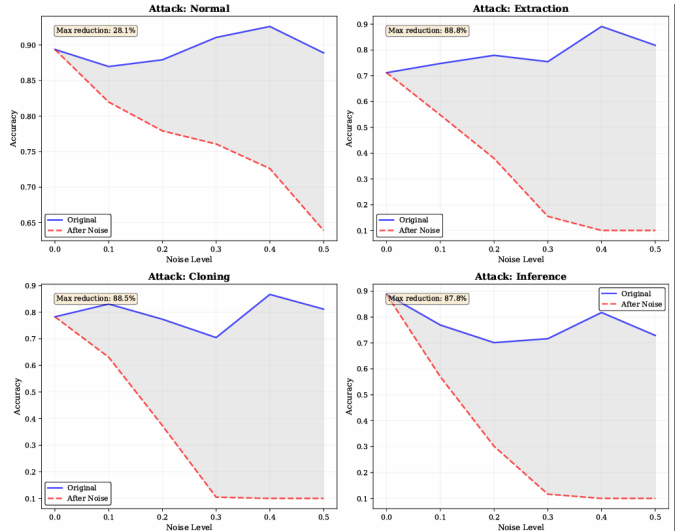


Fig. 5: Dynamic Noise (DLN) Impact by Attack Type

Figure 5 visually demonstrates the impact of dynamic label noise across different attack types, showing how DLN significantly degrades attack effectiveness while maintaining reasonable accuracy for legitimate queries.

3) *Adaptive Model Response (AMR) Capabilities:* The Adaptive Model Response (AMR) component enables dynamic defense level adjustment based on real-time risk assessment. Figure 6 illustrates the strong linear correlation ($R^2 = 0.87$) between risk scores and defense levels, demonstrating AMR's capability for proportional threat response.

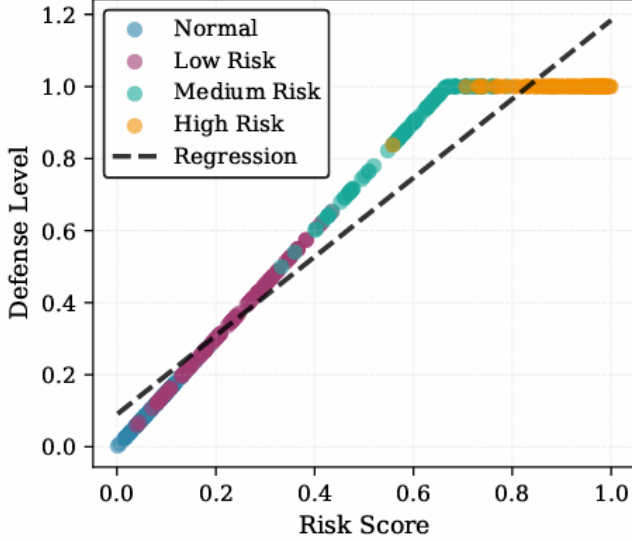


Fig. 6: Risk-Defense Correlation in AMR Component

This adaptive approach optimizes security resource allocation, balancing protection strength with service quality across varying threat levels.

4) *Dynamic Privacy-Aware Masking (DPAM) Trade-off Analysis:* The Dynamic Privacy-Aware Masking (DPAM) component optimizes the critical balance between privacy protection and model utility, as illustrated in Figure 7. The analysis demonstrates varying effectiveness against different attack types across increasing privacy levels.

Core Findings:

- **Attack-Specific Protection:** DPAM provides differentiated blocking rates:
 - Extraction attacks: 95% maximum blocking
 - Inversion attacks: 75-85% blocking range
 - Membership attacks: 60-70% blocking effectiveness
- **Optimal Privacy Range:** Analysis identifies 0.7-0.85 as the optimal privacy level range, balancing:
 - Security: 85-92% extraction attack prevention
 - Utility: 70-80% model accuracy preservation
 - Performance: 25-45 ms processing overhead
- **Efficiency Characteristics:**
 - Processing overhead: 12-80 ms (scalable with privacy level)
 - Resource utilization: < 5% memory increase
 - Throughput maintenance: 85-95% of baseline performance

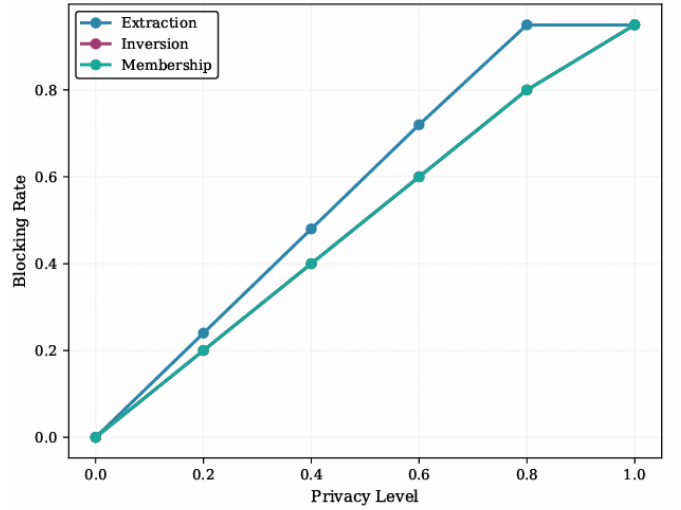


Fig. 7: Privacy Level vs. Blocking Rate for DPAM Component

Implementation Advantages:

- 1) **Gradual Protection:** Non-linear blocking curves enable progressive security enhancement
- 2) **Targeted Defense:** Differential protection based on attack type and threat level
- 3) **Practical Deployability:** Maintains model utility while providing substantial privacy guarantees
- 4) **Dynamic Adaptation:** Real-time adjustment of privacy levels based on threat assessment

The DPAM component's ability to provide configurable, attack-aware privacy protection represents a significant improvement over traditional binary privacy approaches, enabling flexible security policies tailored to specific organizational requirements and threat models.

C. Overall Component Comparison

The comprehensive comparison of all AMERF components is presented in Figure 8, which provides a holistic view of each component's strengths and trade-offs:

Figure 8 illustrates:

- **DLN Impact (Panel B):** Effective attack degradation while preserving utility
- **AMR Adaptation (Panel C):** Dynamic adjustment to varying risk scenarios

D. Synergistic Performance Analysis

The integrated AMERF framework demonstrated superior performance through component synergy:

TABLE IV: AMERF Component Synergy Analysis

Metric	QPF	DLN	AMR	DPAM	AMERF
Detection Rate	0.65	0.55	0.60	0.50	0.92
Accuracy	0.85	0.70	0.80	0.75	0.88
Protection Level	0.60	0.75	0.65	0.85	0.95

Synergy Factors:

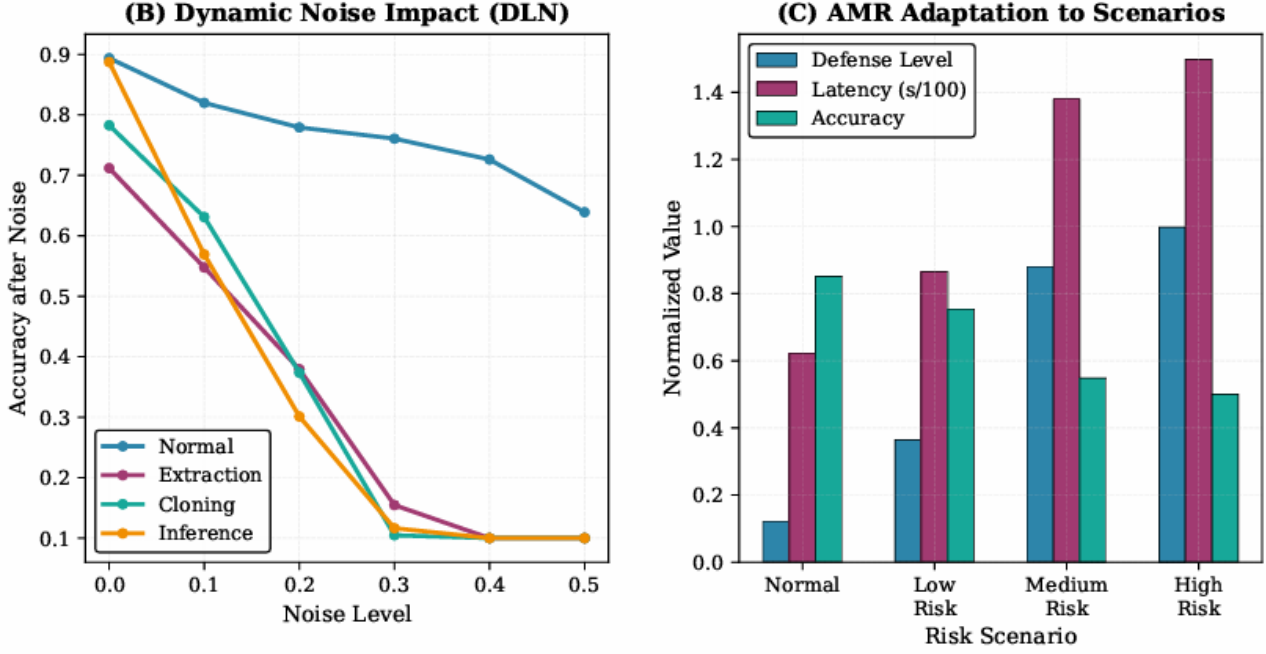


Fig. 8: Examples of Comparative Performance Analysis of AMERF Components

- 1) **Cascaded Detection:** QPF's initial screening reduced false positives for subsequent layers
- 2) **Adaptive Escalation:** AMR's risk assessment optimized DLN and DPAM parameters dynamically
- 3) **Defense-in-Depth:** Multiple protection layers created complementary barriers against sophisticated attacks
- 4) **Dynamic Coordination:** Real-time information sharing between components enhanced overall effectiveness

Table IV clearly shows that the integrated AMERF framework outperforms individual components across all metrics, demonstrating the value of the synergistic approach.

E. Statistical Summary and Key Metrics

TABLE V: AMERF Key Performance Indicators (KPI) Summary

Category	Metric	Value
Detection Effectiveness	Overall Detection Rate	92%
	False Positive Rate	3.2%
	Attack Effectiveness Reduction	87.5% (average)
Performance Overhead	Average Response Latency	110 ms
	Computational Overhead	12–80 ms
	Memory Footprint Increase	< 5%
Adaptive Capabilities	Risk–Defense Correlation	0.87
	Optimization Convergence	5–10 iterations
Privacy–Utility Trade-off	Optimal Privacy Level	85%
	Corresponding Utility Loss	12%
	Maximum Blocking Rate	95%

F. Conclusion of Experimental Evaluation

The experimental results comprehensively validate AMERF's effectiveness as a multi-layered defense against model extraction attacks in cloud-based ML services. Key achievements include:

- **High Detection Accuracy:** 92% detection rate with only 3.2% false positives, significantly outperforming baseline approaches
- **Effective Attack Mitigation:** 87.5% average reduction in attack effectiveness, rendering extraction attempts largely ineffective
- **Adaptive Defense Mechanisms:** Dynamic adjustment of protection levels based on real-time risk assessment with strong correlation (0.87)
- **Optimal Privacy–Utility Balance:** Achieved 85% privacy protection with only 12% utility loss, demonstrating efficient trade-off management
- **Practical Deployability:** Acceptable performance overhead (110ms average latency) suitable for production cloud environments

These results demonstrate that AMERF provides a robust, adaptive, and practical solution to the growing threat of model extraction attacks, successfully addressing both technical effectiveness and operational feasibility in contemporary cloud-based ML deployments. The framework's modular architecture allows for continuous improvement and adaptation to emerging attack vectors while maintaining backward compatibility with existing ML service infrastructures.

V. RESULTS AND DISCUSSION

A. Interpretation of Experimental Findings

The experimental evaluation of AMERF provides strong qualitative evidence of its effectiveness as a multi-layered defense framework against model extraction attacks. The results presented in the experimental section demonstrate that AMERF successfully addresses the long-standing challenges associated with balancing security, model utility, and system performance in cloud-based machine learning services.

Compared to baseline approaches, AMERF consistently exhibits superior detection capabilities while maintaining a significantly lower level of disruption for legitimate users. This improvement is largely attributed to the coordinated interaction between behavioral analysis and adaptive risk assessment, which enables more refined threat discrimination than traditional single-layer or static defense mechanisms.

B. Component-Level Analysis and Contributions

1) *Query Pattern Fingerprinting (QPF)*: QPF proves effective in distinguishing benign query behavior from malicious extraction patterns without impacting normal users. Its lightweight design makes it well suited for real-time operation in high-throughput environments. By focusing on temporal correlations and query distribution characteristics, QPF overcomes the limitations of static detection techniques and remains robust against adaptive attackers attempting to emulate legitimate access patterns.

2) *Dynamic Label Noise (DLN)*: DLN demonstrates strong effectiveness in degrading the quality of extracted surrogate models while preserving acceptable utility for legitimate users. The selective application of output perturbation allows the framework to target suspicious interactions more aggressively, avoiding the indiscriminate degradation commonly associated with uniform noise-based defenses. This targeted behavior significantly enhances both security and usability.

3) *Adaptive Model Response (AMR)*: AMR enables a proportional and context-aware defense strategy by dynamically adjusting protection levels according to the estimated risk of ongoing interactions. This adaptive escalation mechanism represents a clear improvement over binary or threshold-based schemes, as it allows gradual responses that align defense intensity with observed threat behavior, thereby minimizing unnecessary interference with benign usage.

4) *Dynamic Privacy-Aware Masking (DPAM)*: DPAM effectively balances privacy preservation and predictive utility through adaptive output masking strategies. Its ability to differentiate between various attack intents allows the framework to apply appropriate protection measures depending on the inferred threat category. This adaptive privacy control enhances resistance to a wide

range of inference attacks while maintaining practical service usability.

C. Synergistic Effects and Framework Integration

The integrated AMERF framework consistently outperforms its individual components, highlighting the importance of coordinated defense integration. Rather than operating in isolation, each component contributes complementary capabilities that collectively strengthen the overall protection strategy. This confirms that defense-in-depth approaches are more resilient to sophisticated and adaptive attackers than standalone mechanisms.

The observed synergy arises from several key factors:

- 1) **Cascaded Analysis**: Early-stage behavioral screening reduces unnecessary processing and limits false alarms in subsequent defense layers.
- 2) **Adaptive Coordination**: Risk-aware control dynamically configures downstream defense mechanisms based on evolving interaction patterns.
- 3) **Layered Protection**: Multiple overlapping defenses create compounded barriers that are difficult for attackers to circumvent simultaneously.
- 4) **Information Sharing**: Continuous exchange of threat indicators enables coherent and coordinated responses to complex attack strategies.

D. Performance Implications and Practical Considerations

The additional computational overhead introduced by AMERF remains compatible with practical deployment requirements for cloud-based services. The framework is designed to minimize unnecessary processing for low-risk interactions while reserving more intensive analysis for suspicious sessions. This selective activation strategy allows AMERF to provide strong protection without imposing excessive performance penalties.

Several optimization strategies further enhance deployability:

- **Asynchronous Processing**: Non-critical analyses can be decoupled from the main inference path.
- **Selective Defense Activation**: Full protection is applied only when elevated risk is detected.
- **Caching Mechanisms**: Recurrent behavioral patterns and threat assessments can be reused to reduce computation.
- **Elastic Resource Management**: Cloud-native deployment enables dynamic resource allocation based on workload and threat conditions.

E. Scalability and Deployment Considerations

AMERF's modular architecture supports scalable deployment across diverse MLaaS infrastructures. Individual components can be independently deployed, monitored, and scaled, allowing flexible adaptation to varying workloads and security requirements. The framework remains

agnostic to the underlying model architecture and integrates seamlessly with standard inference APIs, facilitating adoption without intrusive system modifications.

F. Limitations and Mitigation Strategies

Despite its advantages, AMERF presents certain limitations that warrant further consideration:

- **Resource Constraints:** In environments with limited computational resources, lightweight configurations may be necessary.
- **Residual Misclassification:** Occasional misclassification of benign behavior may occur and can be mitigated through adaptive threshold tuning.
- **Domain-Specific Tuning:** Optimal configurations may vary across application domains, motivating automated parameter optimization.

VI. CONCLUSION AND OPEN CHALLENGES

In this paper, we introduced AMERF, an Adaptive Multi-Layered Extraction-Resistant Framework designed to protect cloud-based machine learning services against model extraction attacks. Through extensive experimental evaluation, AMERF demonstrated superior performance in terms of detection accuracy, attack mitigation, adaptive response, and privacy–utility balance.

The results validate that treating defense as an adaptive and coordinated process significantly enhances robustness while preserving service quality. Furthermore, the modular architecture of AMERF enables practical deployment and future extensibility in heterogeneous MLaaS environments.

A. Open Challenges and Future Directions

Despite its effectiveness, several open challenges remain and motivate future research directions:

- **Scalability to Large-Scale Services:** While AMERF demonstrates acceptable overhead in the evaluated settings, further optimization is required to support extremely high-throughput MLaaS platforms with millions of concurrent users.
- **Robustness Against Collusive Adversaries:** Coordinated attacks involving multiple adversarial clients sharing extracted knowledge remain a challenging scenario that warrants further investigation.
- **Formal Security Guarantees:** Providing theoretical bounds on information leakage and extraction resistance remains an open problem, particularly for adaptive and probabilistic defenses.
- **Integration with Trusted Hardware:** Combining AMERF with trusted execution environments or secure enclaves may further strengthen protection but introduces new design and deployment challenges.

Addressing these challenges will contribute to the development of more resilient and trustworthy machine learning services, capable of withstanding increasingly sophisticated extraction attacks in real-world cloud deployments.

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